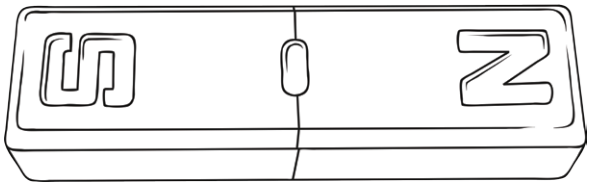
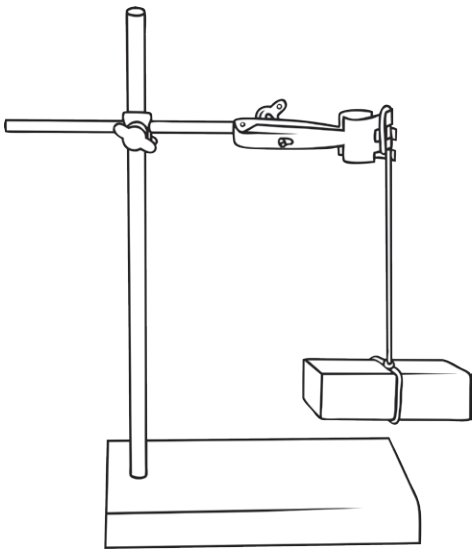
Name:	Class:	Date given:
		Date due in:

Magnetism

1. Add arrows to show the direction of the magnetic field lines around the bar magnet. [2]

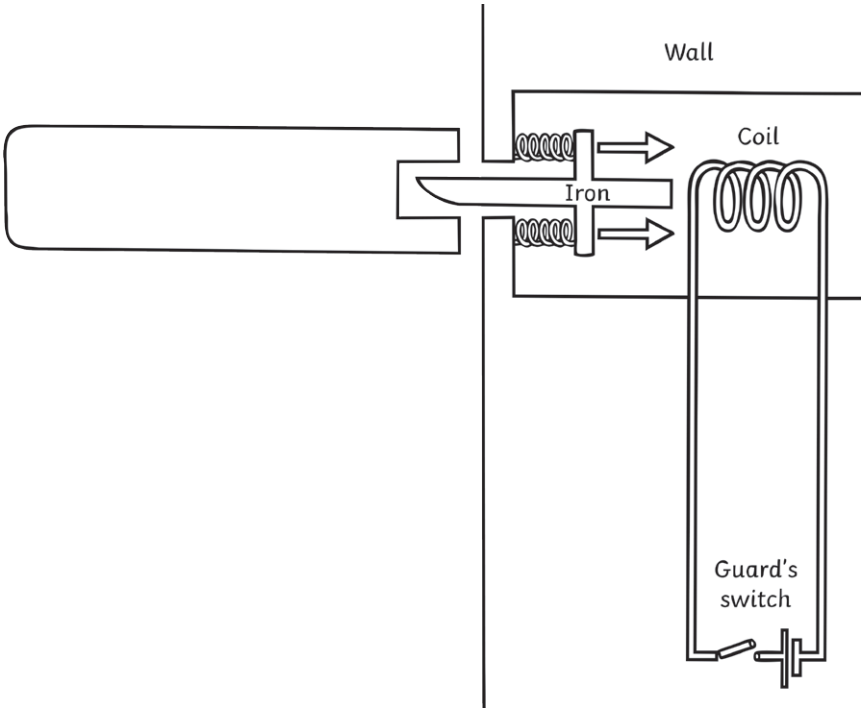


2. Michael is using this apparatus to test different materials and sort them into magnetic and non-magnetic. Explain how he can use this apparatus to do his investigation. [2]



visit twinkl.com

Look at the diagram carefully. It shows a door-locking system.



3a. What is the name of the type of magnet used in this door? [1]

3b. Explain why the iron bolt moves to the right when the switch is pushed. [3]

3c. Circle the other materials which could be used in place of the iron bolt. [2]

copper	nickel	steel
cobalt	aluminium	lead

visit twinkl.com

3d. The guard is concerned that the magnet in the door lock is not strong enough. Suggest two ways that the strength of the magnet could be increased. [2]

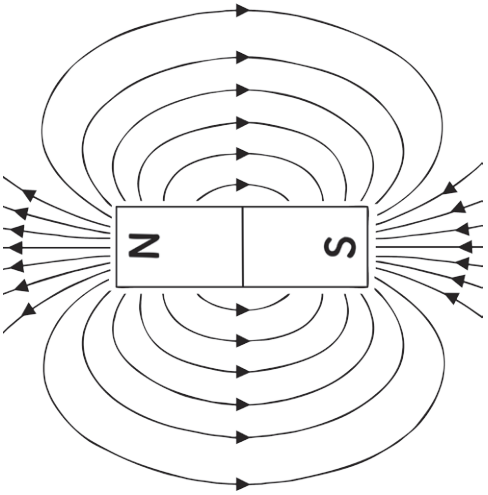
Learning Outcomes (tick if achieved)

Q1	I can draw the electromagnetic field of a bar magnet.	
Q2	I can describe how to test whether materials are magnetic.	
Q3	I can describe and explain how electromagnets work.	

visit twinkl.com

Magnetism **Answers**

1. Add arrows to show the direction of the magnetic field lines around the bar magnet. [2]



2. Michael is using this apparatus to test different materials and sort them into magnetic and non-magnetic. Explain how he can use this apparatus to do his investigation. [2]

Michael could hold the different materials to the magnet if they drop – they are non-magnetic / if they stick to the magnetic – they are magnetic

3a. What is the name of the type of magnet used in this door? [1]

electromagnet

3b. Explain why the iron bolt moves to the left when the switch is pushed. [3]

- **circuit is complete so current flows**
- **current passing the coil creates the magnetic field**
- **iron is magnetic so attracted towards the electromagnet**

3c. Circle the other materials which could be used in place of the iron bolt. [2]

copper	nickel	steel
cobalt	aluminium	lead

3d. The guard is concerned that the magnet in the door lock is not strong enough. Suggest two ways that the strength of the magnet could be increased. [2]

- **increase number of loops in the coil**
- **use a soft iron core within the coil**
- **increase the current in the circuit**